

Serie 9

OPERATIONS ON SUBSPACES AND LINEAR MAPS

1. Linear complement. Let F be a field. Find a linear complement of the following subspaces in $M_{n \times n}(F)$ (see Serie 8 for definitions):

- (a) The subspace of upper triangular matrices;
- (b) The subspace of symmetric matrices.

2. Linear map. Let $b, c \in \mathbb{R}$ and let $\mathbb{R}[x]$ denote the space of polynomial functions of 1 variable with coefficients in $\mathbb{R}$. Define $T : \mathbb{R}[x] \rightarrow \mathbb{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^2 x^3 p(x) dx + cp(0)^2 \right).$$

Show that T is linear if and only if $b = c = 0$.

3. Intersection of subspaces. Suppose that U and V are both 4-dimensional subspaces of $\mathbb{C}^6$, viewed as a vector space over $\mathbb{C}$. Prove that there exists 2 vectors in $U \cap V$ such that neither of these vectors is a scalar multiple of the other.

4. Dimension. Suppose that $\{v_1, \dots, v_m\}$ is linearly independent in V and let $w \in V$. Prove that

$$\dim \text{Sp}(v_1 + w, v_2 + w, \dots, v_m + w) \geq m - 1.$$

5. Sum of subspaces. Let V be a vector space over a field F and consider 3 linear subspaces U_1, U_2, U_3 such that $V = U_1 + U_2 + U_3$ and for $i, j \in \{1, 2, 3\}$ such that $i \neq j$, we have $U_i \cap U_j = \{0\}$.

Is it true that for any $v \in U_1 + U_2 + U_3$ there exists a unique triple (u_1, u_2, u_3) with $u_i \in U_i$ such that $v = u_1 + u_2 + u_3$?

6. Nested subspaces. Let V be a finite-dimensional vector space over a field F and let

$$V \supseteq U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots \supseteq U_k \supseteq \cdots$$

be an infinite sequence of nested subspaces.

- (a) Show that this sequence stabilises, i.e. show that there exists a $N \in \mathbb{N}$ such that for all $n \geq N : U_n = U_N$.
- (b) Is this still the case if we now assume that V is infinite-dimensional?
- (c) Assume that V is infinite-dimensional and that $\dim U_n \geq 1$ for all $n \in \mathbb{N}$. What can you say about $\bigcap_{n \in \mathbb{N}} U_n$?

7. Dimension and basis. Consider the following subspace of K^n :

$$U := \left\{ (\alpha_1, \dots, \alpha_n) \in K^n \mid \sum_{i=1}^n \alpha_i = 0 \right\},$$

$$D := \{(\alpha, \dots, \alpha) \in K^n \mid \alpha \in K\}.$$

Determine a basis and compute the dimension of $U, D, U \cap D$, and $U + D$.

Remark: Do not forget to consider the case where K is a field such that $n \cdot 1 = 0$.

Multiple Choice Fragen. More than one answer can be correct.

Frage 1. Which of the following maps are linear?

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x + y, 2x, 0)$
- $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x + y, 2xy, 0)$
- $f : K^3 \rightarrow K^2, (x, y, z) \mapsto (\alpha x + \beta y + \gamma z, \delta x + \varepsilon y + \eta z)$ for fixed $\alpha, \beta, \gamma, \delta, \varepsilon, \eta$ in the field K

Frage 2. Which of the linear maps below can be written as $x \mapsto Ax$, where A is the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 2 \end{pmatrix}$$

- $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, f(x, y) = (2x + y, x, 2y)$
- $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, f(x, y, z) = (2x + y, x + 2z)$